

TickExp: A Unified Multi-Service Booking Application

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BONAFIDE CERTIFICATE

Certified that this Project titled **TickExp: A Unified Multi-Service Booking Application** is the bonafide work of **Kumaran R (230701158)** and **Lingeswaran V (230701163)** who carried out the work under my supervision. Certified further that to the best of my knowledge, the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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INTERNAL EXAMINER

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ABSTRACT

TickExp is a unified multi-service booking application designed to provide a seamless platform for booking movies, trains, and buses within a single system. The application addresses the limitations of using multiple independent platforms by integrating diverse booking services into one cohesive ecosystem, thereby improving user convenience and accessibility. It enables users to browse available options, view detailed information, and perform bookings efficiently without switching between applications.

The system incorporates real-time data integration from multiple backend sources to ensure up-to-date information for movies, train schedules, and bus availability. It also includes location-based personalization to display relevant options based on the user's region, enhancing the overall user experience. A secure digital wallet and payment mechanism are implemented to handle transactions, along with a reward system that encourages user engagement.

In addition, TickExp generates digital tickets with QR codes and supports PDF export for easy access, sharing, and verification. The application is built using modern technologies and follows a Clean Architecture approach to ensure scalability, modularity, and high performance. Overall, TickExp provides a reliable, efficient, and future-ready solution by combining transportation and entertainment booking services into a single integrated platform.

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CHAPTER 1

INTRODUCTION

1.1 General

TickExp is a unified and interactive booking platform designed to provide a seamless experience across multiple services, including movies, trains, and buses, within a single application. It allows users to explore options, check availability, and perform bookings without switching between different platforms. The system focuses on personalization by adapting content based on user location and preferences while simulating real-world booking features such as dynamic availability, secure payments, digital ticket generation, and reward mechanisms. By integrating multiple booking domains into one ecosystem, TickExp simplifies the user journey and delivers a consistent, efficient, and reliable experience. It also enhances user convenience by centralizing all booking activities in one place. The platform is designed to handle diverse use cases while maintaining a smooth and intuitive flow. Overall, it aims to provide a scalable solution that improves accessibility and user engagement.

1.2 Scope of the Work

TickExp focuses on developing a unified booking platform that allows users to access and manage multiple services such as movie, train, and bus bookings within a single application. The system is designed to simplify the overall booking process by providing a centralized interface where users can explore options, check availability, and complete transactions efficiently.

- Browse movies, train schedules, and bus services
- View details such as timings, availability, and routes
- Select seats and perform bookings
- Make payments through a digital wallet system

- Earn reward points for transactions
- Generate and access digital tickets

The scope also includes integrating real-time data sources, location-based personalization, and secure transaction handling to simulate real-world booking systems. Overall, the project ensures a smooth, consistent, and user-friendly experience across different booking domains while maintaining performance and reliability.

1.3 Aim and Objectives of the Project

The aim of the TickExp project is to develop a unified, scalable, and user-friendly booking platform that integrates movie, train, and bus services into a single application, providing a seamless and personalized user experience. The project focuses on enabling users to explore options, check real-time availability, and complete bookings efficiently by leveraging dynamic data from multiple backend sources, including TMDB, RapidAPI-based IRCTC services, and a simulated RedBus API routed through a Vercel proxy. It incorporates location-based personalization using FusedLocationProvider and geocoding, along with a secure payment system powered by the Razorpay SDK and a digital wallet mechanism. The system also includes QR-based ticket generation using ZXing and PDF export functionality, while following Clean Architecture principles with Jetpack Compose, ViewModels, Retrofit, and Kotlin Coroutines to ensure modularity, asynchronous processing, and high performance. Overall, the objective is to build a robust and extensible multi-service booking ecosystem that delivers real-world functionality, efficient data handling, and a smooth, consistent user experience across all modules. It also emphasizes secure data handling, transaction integrity, and efficient state management across different services. The platform is designed to support future scalability, allowing easy integration of additional booking domains and advanced features.

CHAPTER 2

LITERATURE SURVEY

Research on unified booking platforms highlights the growing need to integrate multiple services such as transportation and entertainment into a single system to enhance user convenience. Existing studies emphasize challenges related to data integration, real-time availability, and scalable system design, which are critical factors in developing platforms similar to TickExp [1], [2].

Unified platforms aim to consolidate multiple services into one interface, reducing the dependency on separate applications. Studies show that systems combining services like bus and hotel reservations use full-stack architectures and RESTful APIs to provide real-time availability and secure transactions. Similarly, integrated rail and bus booking systems enable seamless online ticketing and eliminate traditional queue-based processes [3], [7].

Multi-service integration addresses the issue of fragmented user experiences in travel planning. Research indicates that centralized systems combining multiple domains such as bus, train, and other services improve efficiency through unified databases and administrative control. Multimodal aggregator applications further enhance user convenience by incorporating location-based services to provide optimized travel options [4], [6].

From an architectural perspective, scalable and modular designs are essential for such platforms. Clean Architecture principles are widely adopted to maintain separation of concerns and support extensibility. Studies on large-scale booking systems demonstrate the use of microservices, load balancing, and distributed systems to handle high user demand and ensure system reliability [3], [8].

Payment processing and security remain key components in unified booking systems. Literature highlights the importance of secure payment gateways such as Razorpay and PayPal, which ensure transaction integrity and user trust.

Additionally, verification mechanisms and secure endpoints are necessary to prevent fraud and maintain data security across multiple services [5], [9].

Personalization and location-based features play a significant role in enhancing user satisfaction in integrated platforms. Research shows that applications leveraging GPS-based services improve multimodal planning by suggesting nearby options and relevant services, reducing user effort and decision complexity. Tourism-based systems further utilize user preferences and history to deliver dynamic and context-aware recommendations, aligning with modern booking applications [4], [8].

Ticket management innovations such as QR code generation and PDF-based ticketing improve accessibility and verification in unified systems. Studies highlight the use of barcode technologies for quick validation, reducing fraud and supporting secure digital transactions. These mechanisms also enable offline access and interoperability with external validation systems, which are essential for scalable real-world deployment [5], [7].

Overall, the literature suggests that unified booking platforms are evolving toward highly integrated, scalable, and secure ecosystems that combine multiple services into a single application. These advancements provide a strong foundation for developing systems like TickExp, which aim to deliver convenience, efficiency, and a seamless user experience [10].

CHAPTER 3

SYSTEM DESIGN

3.1 ARCHITECTURE DIAGRAM

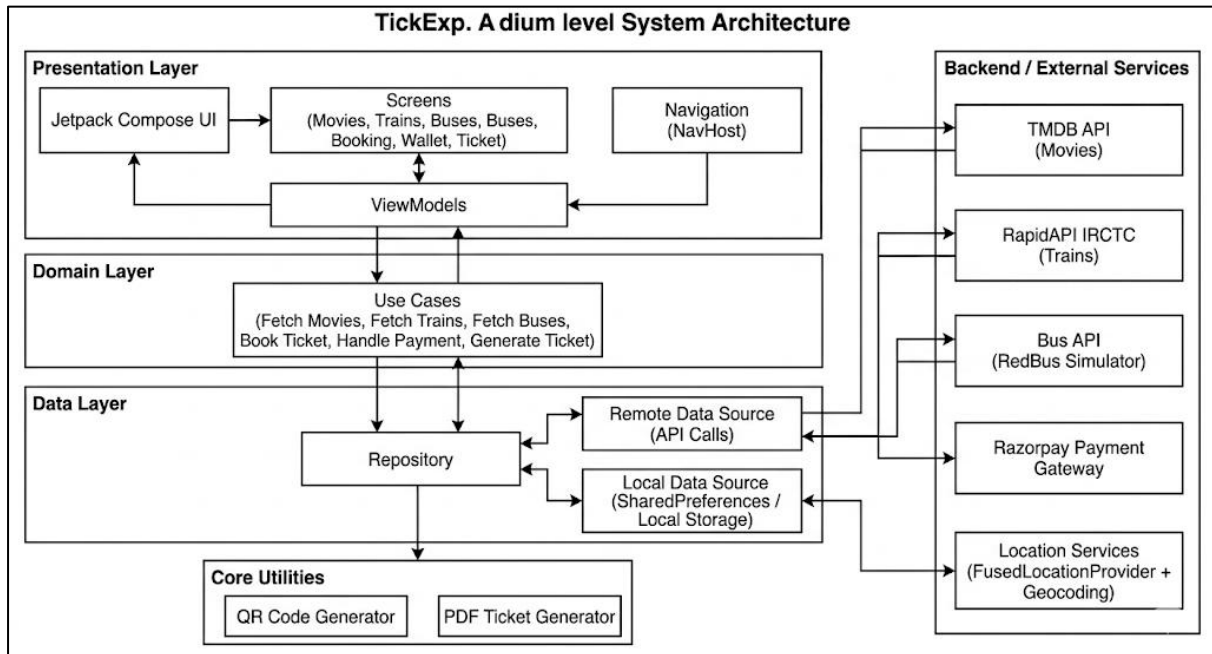


Fig 3.1.1 Architecture Diagram

TickExp follows a layered Clean Architecture approach to support its multi-service booking system for movies, trains, and buses, ensuring scalability and maintainability. The system is divided into three layers: Presentation, Domain, and Data. The Presentation Layer handles the user interface, navigation, and interactions through screens and ViewModels. The Domain Layer manages core business logic such as fetching data, processing bookings, and handling payments across different services. The Data Layer manages repositories that coordinate with multiple backend sources for API communication and local storage. The architecture integrates external services through a backend proxy to provide real-time data, secure payment processing, and location-based personalization. It also uses efficient asynchronous operations to maintain smooth performance. Additionally, components like QR code generation and PDF ticket creation support secure ticket management and sharing. Overall, the system is modular,

efficient, and extensible, ensuring a consistent user experience across all booking services.

3.2 USE CASE DIAGRAM

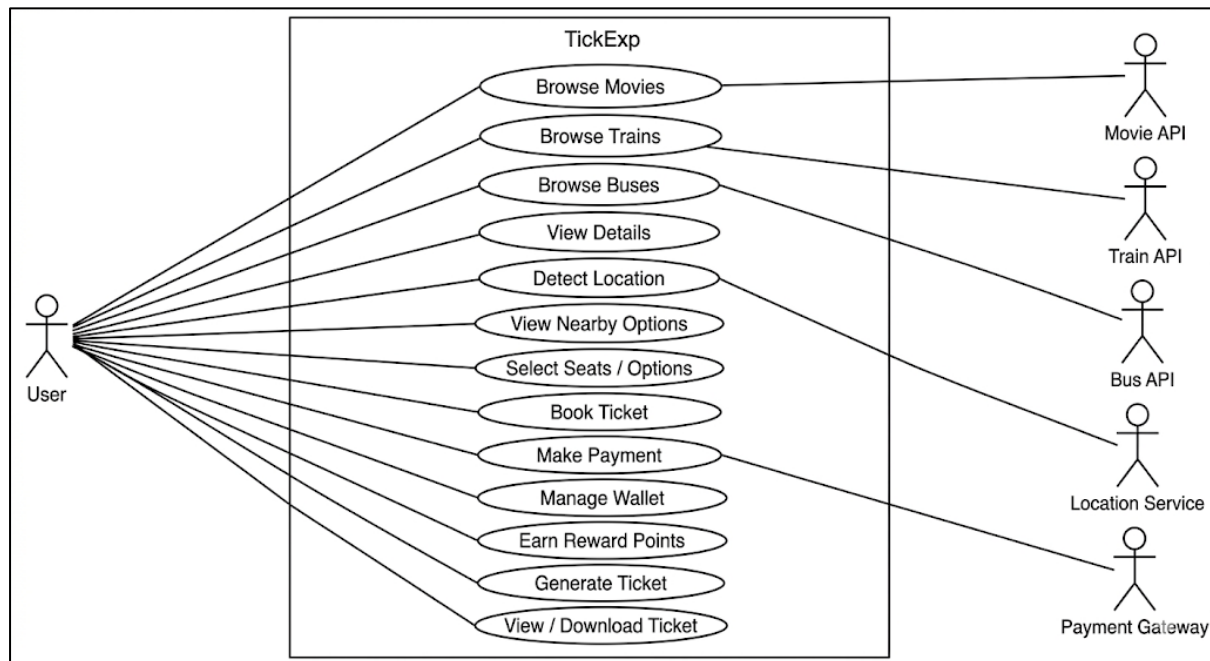


Fig 3.2.1 Use Case Diagram

The use case diagram of TickExp illustrates how the user interacts with a unified platform to access multiple services such as movies, trains, and buses. The user can browse available options, view details, detect location, and explore relevant theatres, train schedules, or bus routes based on preferences. The system allows users to select options, choose seats where applicable, and complete bookings through a simple and guided process. It also connects with external services to fetch real-time data, provide location-based personalization, and handle secure payments. After successful booking, the system generates a digital ticket that can be viewed or downloaded, ensuring a smooth and integrated booking experience across all services. The system also maintains user history and transaction details to support easy access and future bookings.

3.3 HARDWARE SPECIFICATIONS

The TickExp application is designed to run efficiently on modern computing systems and Android mobile devices, ensuring smooth performance across its multi-service booking features.

DEVELOPMENT SYSTEM REQUIREMENTS

- Processor: Intel Core i5 or higher
- RAM: Minimum 8 GB
- Storage: Minimum 20 GB free space
- Operating System: Windows 10 / macOS / Linux
- Development Tools: Android Studio

MOBILE DEVICE REQUIREMENTS

- Android Version: Android 8.0 (Oreo) or above
- RAM: Minimum 4 GB
- Storage: Minimum 150 MB free space
- Internet Connectivity: Required for real-time data and API Endpoints
- GPS Sensor: Required for location-based personalization

SERVER REQUIREMENTS

- Backend: Vercel (Cloud-based Proxy Server)
- API Integration: TMDB API, RapidAPI (IRCTC), Custom Bus API
- Network: Stable internet connection for API routing and data fetching

These hardware requirements ensure efficient performance, real-time data processing, and a seamless user experience across all supported booking services.

3.4 SOFTWARE SPECIFICATIONS

The TickExp application is developed using modern software technologies to ensure high performance, scalability, and a seamless multi-service booking experience. The software stack includes frontend development frameworks, backend services, API integrations, and supporting tools.

Frontend Software Requirements

- Programming Language: Kotlin
- UI Framework: Jetpack Compose
- Architecture: Clean Architecture with MVVM pattern
- Navigation: Navigation Compose (Multi-stack NavHost)
- Design System: Material Design 3 with Glassmorphism UI
- Development Environment: Android Studio

The frontend is responsible for rendering dynamic user interfaces, handling navigation, managing UI states, and providing smooth interactions across movie, train, and bus booking modules.

Backend Software Requirements

- Backend Platform: Vercel (Proxy Server)
- API Type: RESTful APIs
- API Integration: TMDB API, RapidAPI (IRCTC), Custom Bus API
- Communication: Retrofit 2 with Kotlin Coroutines

The backend layer handles API routing, secure communication, and real-time data fetching from multiple external services.

Database Requirements

- Local Storage: SharedPreferences
- Data Type: Key-value storage for user data (profile, wallet, tickets)

The system stores essential user information locally to ensure persistence and quick access.

Third-Party APIs and Services

- Movie Data: TMDB API
- Train Services: RapidAPI (IRCTC endpoints)
- Bus Services: Custom RedBus Simulator API
- Payment Gateway: Razorpay SDK
- Location Services: Google FusedLocationProvider & Geocoding

These services provide real-time data, payment processing, and location-based personalization.

Version Control

- Platform: Git
- Repository Hosting: GitHub

Version control is used to manage source code, track changes, and support collaborative development.

Operating System Requirements

- Development: Windows / macOS / Linux
- Deployment: Android OS (Version 8.0 and above)

Other Supporting Tools

- Build System: Gradle
- API Testing Tools: Postman (optional)
- Emulator: Android Emulator / Physical Device

These software components ensure that TickExp operates efficiently, supports real-time multi-service integration, and delivers a smooth and reliable user experience.

CHAPTER 4

PROPOSED SYSTEM

The proposed system, TickExp, is a unified and intelligent booking platform designed to integrate multiple services such as movie ticket booking, train reservations, and bus ticketing into a single cohesive application. The primary goal of the system is to eliminate the need for users to switch between multiple apps by providing a centralized platform where they can explore, compare, and book services efficiently. The system emphasizes user convenience, personalization, and real-time interaction by delivering dynamic content such as currently running movies, live train schedules, and available bus routes. By combining multiple domains into one ecosystem, TickExp aims to simplify the overall booking process while ensuring a consistent and seamless user experience.

From a technical perspective, the system is built using modern Android development practices with Jetpack Compose for UI design and Clean Architecture to ensure scalability, modularity, and maintainability. The application follows a layered structure consisting of Presentation, Domain, and Data layers, where each layer handles specific responsibilities such as UI rendering, business logic, and data management. The use of ViewModels ensures efficient state management and smooth interaction between the UI and backend logic, while Navigation is handled through a multi-stack NavHost to support multiple booking flows.

The system integrates multiple backend engines through a secure Vercel proxy server, enabling real-time data fetching from different sources. Movie data is retrieved from the TMDB API, train-related information such as schedules and PNR status is accessed through RapidAPI-based IRCTC services, and bus data is provided through a custom-built RedBus simulator API designed to mimic real-

world systems. These integrations are handled using Retrofit 2 and Kotlin Coroutines, allowing asynchronous operations and ensuring a responsive user interface without delays.

To enhance personalization, the system incorporates a Geo-Velocity Location System that uses FusedLocationProvider and geocoding techniques to detect the user's location and convert it into meaningful information such as city names. This enables the application to dynamically filter and display relevant theatres, train stations, and bus routes based on the user's region. The system also adapts content in real time, improving user engagement and reducing the effort required to find suitable options.

The financial infrastructure of TickExp is designed to simulate a real-world transaction system using the Razorpay SDK. The application includes a digital wallet that allows users to add funds, make payments, and manage their balance efficiently, along with a reward system called Velocity Points that provides incentives for each successful booking.

In terms of ticket management and security, the system generates unique QR codes for each booking using ZXing-based encoding techniques, ensuring that every ticket is secure and verifiable. Additionally, a PDF export engine is implemented to generate digital tickets that can be stored, shared, or accessed easily, while user details are maintained through local persistence for consistency.

The user interface of TickExp is designed using modern UI principles such as glassmorphism, providing a visually appealing and smooth experience. Overall, the proposed system combines real-time data integration, secure transactions, location-based personalization, and efficient architecture to create a scalable, high-performance, and user-friendly multi-service booking platform.

CHAPTER 5

MODULE DESCRIPTION

The TickExp system is divided into multiple functional modules, each responsible for handling a specific part of the unified booking platform. These modules work together to provide a seamless experience across movie, train, and bus booking services.

5.1 Movie Booking Module

This module allows users to browse currently available movies, view details such as ratings and descriptions, and book tickets by selecting theatres, showtimes, and seats. It fetches real-time data and provides an interactive booking flow.

5.2 Train Booking Module

The train module enables users to search for train schedules, check availability, and view PNR status. It provides essential travel details and allows users to proceed with booking in a structured and user-friendly manner.

5.3 Bus Booking Module

This module handles bus search and booking functionalities, including viewing routes, timings, and seat availability. It simulates real-time bus inventory and allows users to select and confirm bookings easily.

5.4 Location Module

The location module detects the user's current position and converts it into a readable location. It helps in filtering nearby theatres, stations, and routes, ensuring personalized and relevant results.

5.5 Payment & Wallet Module

This module manages all financial transactions within the application. It includes a digital wallet for adding funds, making payments, and tracking balances, along with a reward system that provides points for each successful booking.

5.6 Ticket Management Module

This module generates and manages digital tickets for all bookings. It creates secure QR codes and allows users to view, download, and share tickets in PDF format.

5.7 User Management Module

This module handles user-related data such as profile details and preferences. It ensures data persistence and maintains consistency across sessions.

5.8 Navigation & UI Module

This module manages the overall user interface and navigation flow of the application. It ensures smooth transitions between screens and provides a consistent and visually appealing user experience.

CHAPTER 6

OUTPUT

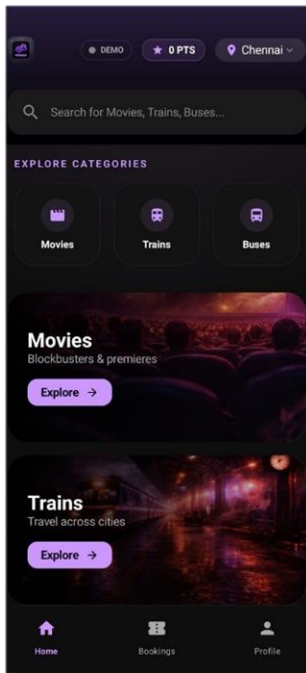


Fig.6.1: Home Screen

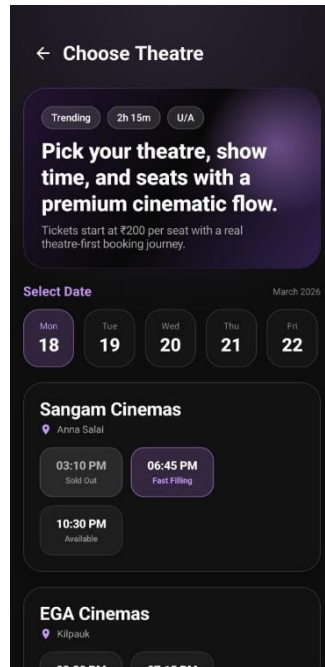


Fig.6.2: Movie Booking Screen

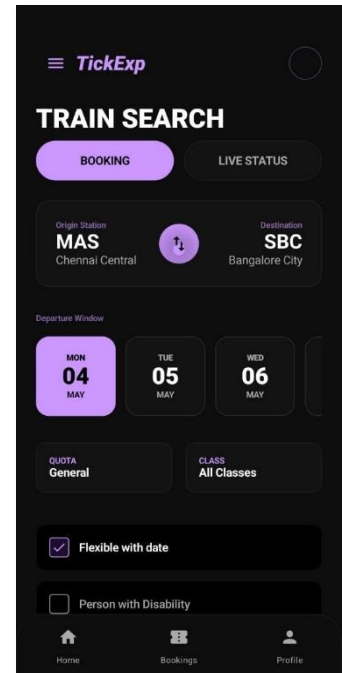


Fig.6.3: Train Booking Screen

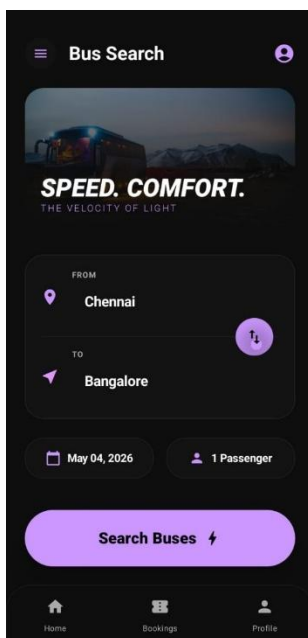


Fig.6.4: Bus Booking Screen

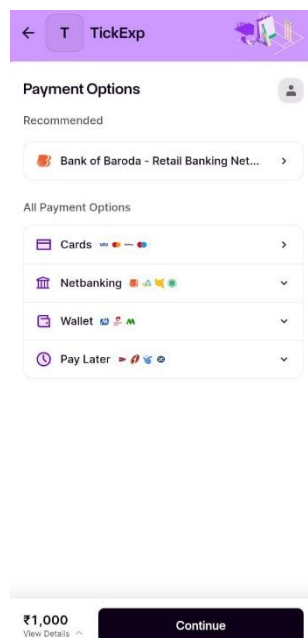


Fig.6.5: Payment Screen

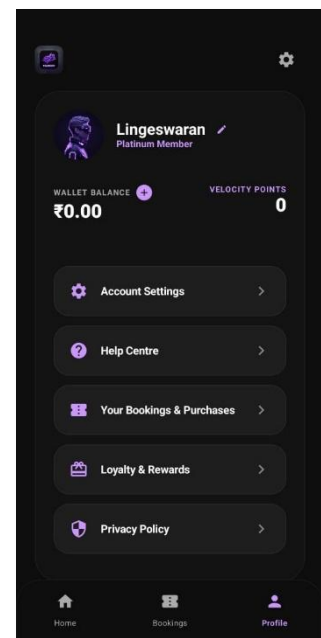


Fig.6.6: Profile Screen

CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENTS

Conclusion:

TickExp presents a comprehensive and unified solution for modern booking needs by integrating movie, train, and bus services into a single platform. The system successfully simplifies the user journey by eliminating the need for multiple applications, offering a centralized interface for exploring options, checking availability, and completing bookings efficiently. With features such as real-time data integration, location-based personalization, secure payment handling, and digital ticket generation, the application closely simulates real-world booking systems.

From a technical perspective, the use of modern technologies and a Clean Architecture approach ensures that the system is scalable, maintainable, and high-performing. The modular design allows easy expansion and integration of additional services in the future. Overall, TickExp demonstrates how multiple domains can be effectively combined into a single ecosystem, delivering convenience, reliability, and an enhanced user experience while laying a strong foundation for future advancements in unified booking platforms.

Furthermore, the project highlights the importance of integrating user-centric design with efficient backend systems to achieve a balanced and robust application. It also opens opportunities for future enhancements such as advanced analytics, real-time seat tracking, and integration of additional services, making TickExp a flexible and future-ready solution in the domain of digital booking systems.

Future Enhancements:

The TickExp system can be further improved by introducing advanced features and expanding its capabilities to enhance user experience and system performance. One of the key enhancements includes implementing real-time seat availability and dynamic pricing for movies, trains, and buses to provide more accurate and flexible booking options. The system can also integrate additional services such as flight booking, hotel reservations, and event ticketing to evolve into a complete travel and entertainment ecosystem.

Further improvements can include the use of artificial intelligence and machine learning for personalized recommendations based on user behavior, preferences, and booking history. Advanced analytics dashboards can be added to provide insights into user activity and system performance. Enhancing security with biometric authentication and stronger encryption mechanisms can improve data protection and transaction safety. Additionally, features like offline ticket access, multi-language support, push notifications for updates, and real-time tracking of bookings can significantly improve usability. Overall, these enhancements will make TickExp more intelligent, scalable, and adaptable to future technological advancements and user needs.

CHAPTER 8

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